

FoldShield⁺⁺ Structural Analysis Report

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Input Structures

Reference PDB	1QLP.pdb
Query PDB	1H8H.pdb
Mode	ca
Ref chain	A
Query chain	A

Classical Coordinate Metrics

These metrics saturate near 1.0 even when function diverges. Use FoldShield++ signals below for structural perturbation insight.

Metric	Value	Note
TM-score	0.300	>=0.50 same fold · >=0.90 nearly identical
RMSD (A)	6.200	Lower is better · >3.0 A = significant deviation

FoldShield++ Signal Scores — Phase 1 + Phase 2

All active signals with scores, fusion weights, and interpretation. Phase 2 signals are highlighted in blue.

Braid similarity	0.827	0.20	strong braid similarity
Motif similarity (UL-DSL)	0.597	0.37	moderate motif similarity
SSEF entropy (Phase 2) [P2]	0.642	0.20	moderate grammar rewrite — conformational transition regime
Persistent homology (Phase 3) [P2]	0.689	0.23	partial persistence similarity

Fusion Layer v2 — Combined Score

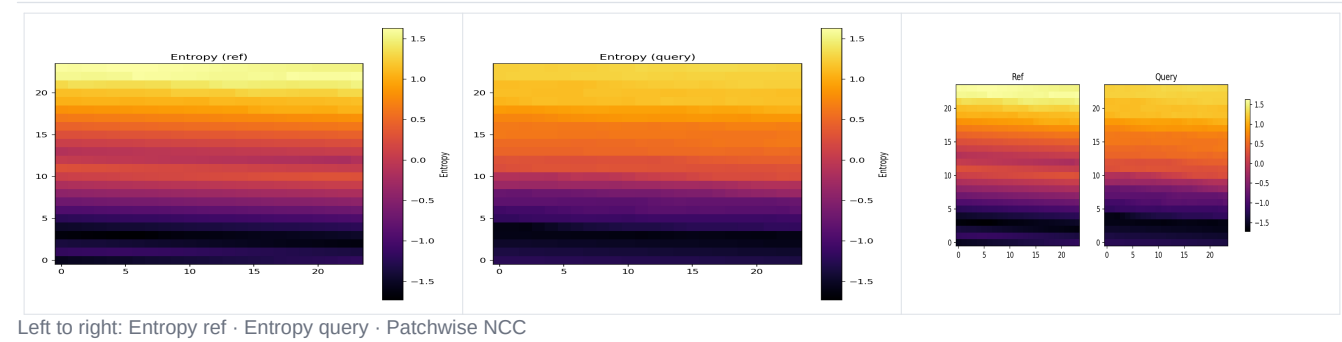
0.6738	phase2	moderately similar protein structures

Phase 2 — Perturbation Classification

SSEF score	Local topology	Gap score	Perturbation class
0.642	0.923	-0.281	Fold-switching protein

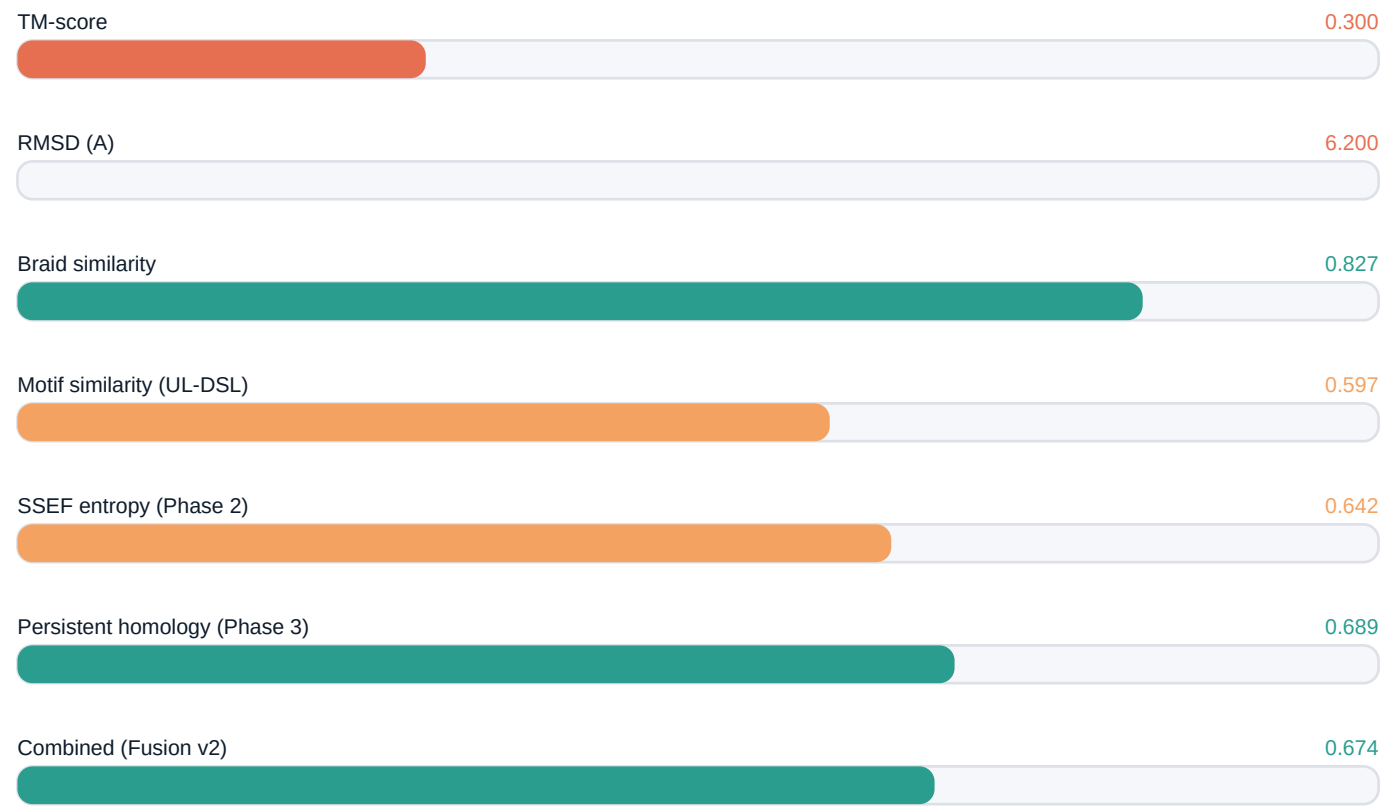
Gap = SSEF minus local topology. <0.15 buried-core mutation | 0.15-0.36 fold-switch | >0.36 conformational transition

Entropy Maps & Signal Profiles



Signal Score Bars

Visual 0-1 scale for all active signals. Green ≥ 0.66 / Amber 0.33-0.66 / Red < 0.33 .



Symbolic Score Detail

Symbolic Similarity Score	0.827	overall nearly identical
Behavioural Correlation Similarity	0.953	entropy maps almost identical (strong dynamic match)
Structural Topology Similarity	0.765	strong generator overlap
Structural Topology Ordering Similarity	0.649	moderate pattern similarity

FoldShield++ Phase 2 | RexCrux Research Laboratory | Signals: Braid / UL-DSL / Global Topology / Local Topology / SSEF Entropy | Fusion: Ridge Regression weighted linear combination

Signal Analysis — Figures

